

AI-Driven Financial Risk Analytics

Interim Report

Kiran Bobba

Walsh College

QM640: Data Analytics Capstone

Dr. Sanhita Karmakar

Term 3

May 10th, 2026

GitHub Repository

The data files, notebooks, modeling artifacts, and reproducible code are available at:

https://github.com/KiranKBobba/Fin_Risk_Analytics.git

The repository is referenced throughout this interim report as the working source for raw data access, intermediate preprocessing outputs, notebook execution, and model experimentation evidence.

Introduction

Background and Context

Financial institutions operate in an environment where digital transaction volume, online lending, automated credit decisioning, and consumer complaint activity continue to grow. These changes create opportunities for efficiency, but they also increase exposure to fraud, credit default, operational anomalies, and consumer harm. Traditional fraud detection systems often depend on fixed business rules and manually tuned thresholds. While these rules are understandable, they are brittle when fraud patterns evolve or when customer behavior changes rapidly. Modern analytics therefore requires adaptive modeling that can detect nonlinear risk patterns, handle high-dimensional data, and provide explainable outputs for business users.

This interim report focuses on two closely related financial analytics problems: fraud/anomaly risk detection and loan eligibility prediction. The first problem asks whether modern deep learning techniques can improve fraud detection performance compared with baseline approaches. The second asks whether counterfactual reasoning can explain model predictions by identifying the minimal financial behavior changes that would alter a prediction. Together, these problems address the central tension in financial AI: institutions want better predictive performance, but they also need explainability, fairness, and governance evidence.

The analysis uses two uploaded notebooks as evidence. The first notebook develops a loan eligibility modeling pipeline using the German Credit dataset enriched with borrower-level micro-features, synthetic time indexing, macro-financial indicators, PCA, XGBoost, rule-based baselines, counterfactual analysis, and fairness tests. The second notebook extends the work toward a public-data fraud/anomaly risk workflow using CFPB complaint data and enhanced

fraud label engineering. The project GitHub repository is provided so an evaluator can inspect the code, data paths, and reproducibility artifacts.

The stakeholders for this project include risk officers, fraud analysts, loan officers, compliance teams, data scientists, model governance teams, consumers, and executive leaders. For a financial institution, the operational relevance is direct: improved fraud detection can reduce loss exposure; improved loan eligibility modeling can improve underwriting consistency; and explainable AI can support audit readiness, customer communication, and responsible AI governance.

Problem Statement

Financial institutions especially Banks and lending institutions such as Fannie Mae and Freddie Mac increasingly depend on algorithmic models to detect fraud/anomaly risk and to support loan eligibility decisions. However, traditional rule-based systems may miss nonlinear or evolving patterns, while advanced machine learning systems can be difficult to explain. This creates a gap between predictive accuracy and decision transparency. The project addresses this gap by comparing baseline methods with machine learning and deep learning approaches, then applying counterfactual analysis to identify which financial behavior variables influence model decisions and how predictions could change under plausible alternative conditions.

Purpose of the Study

The purpose of the study is to build and evaluate a financial risk analytics framework that predicts or classifies risk outcomes while explaining model drivers. The interim work specifically examines (a) baseline and machine learning performance for loan eligibility and fraud/anomaly risk, (b) dimensionality reduction through PCA, (c) engineered borrower and complaint behavior variables, and (d) counterfactual explanations that translate model outputs

into actionable recommendations. The project ultimately aims to support accurate, interpretable, and governance-ready AI use in financial decision-making.

Interim Project Status (Progress Snapshot)

- Completed: German Credit dataset was loaded and enriched with borrower-level micro-features such as loan-to-duration, loan-to-age, installment burden, young borrower indicator, high-amount indicator, and interaction terms.
- Completed: Macro-financial and CFPB-related contextual features were integrated in the first notebook to test whether broader credit conditions can support loan risk modeling.
- Completed: Data cleaning, categorical encoding, normalization, PCA, XGBoost modeling, rule-based baseline modeling, and counterfactual analysis were executed for the loan eligibility workflow.
- Completed: CFPB complaint-based fraud/anomaly risk workflow was initiated with memory-safe sampling, selected features, and improved label engineering logic.
- In progress: Fraud/anomaly risk labels are being refined using randomized mixed-issue CFPB records rather than first-row chronological sampling to avoid single-class labels.
- In progress: Deep learning models for fraud/anomaly detection, including autoencoders and transformer-inspired approaches, are being stabilized after data-label correction.
- Pending: Final fraud/anomaly model comparison, SHAP explainability, additional hyperparameter tuning, and final report recommendations.

Scope and Objectives

The project scope covers financial risk analytics in two related contexts: fraud/anomaly detection and loan eligibility prediction. The interim analysis emphasizes model readiness, interpretability, and experimental evidence rather than a final production deployment. The research questions are listed below with the hypotheses that guide the remaining final-report analysis.

Table 1

Research questions and hypotheses

RQ	ResearchQuestion	Null Hypothesis	AlternativeHypothesis	Interim Evidence
RQ1	How effective are deep learning models in fraud detection?	H0: Deep learning models do not significantly improve fraud/anomaly detection accuracy compared with baseline or traditional models.	Ha: Deep learning models significantly improve fraud/anomaly detection accuracy compared with baseline or traditional models.	Fraud/anomaly workflow; autoencoder and transformer testing in progress.

RQ2	How can counterfactual analysis explain fraud predictions?	H0: Counterfactual explanations do not significantly improve interpretability of fraud/anomaly predictions.	Ha: Counterfactual explanations significantly improve interpretability of fraud/anomaly predictions.	Counterfactual logic identifies feature changes that alter predictions.
RQ3	What is the predictive accuracy of binary classifiers in determining loan eligibility?	H0: Binary classifiers do not provide meaningful predictive performance for loan eligibility.	Ha: Binary classifiers provide meaningful predictive performance for loan eligibility.	XGBoost and rule-based baseline results available.
RQ4	How does causal or counterfactual reasoning improve fairness and transparency in loan eligibility models?	H0: Counterfactual analysis does not improve transparency or fairness evidence.	Ha: Counterfactual analysis improves transparency and fairness evidence.	Counterfactual results, fairness tests, and recommendation analysis available.

The research problems identified are therefore both technical and governance-oriented. Technically, the project asks whether model performance improves when richer feature engineering, PCA, and machine learning are introduced. From a governance perspective, it asks whether counterfactual analysis can convert opaque predictions into explanations that business users can evaluate and act on.

Literature Survey

Literature Review Approach

The literature review was organized around four themes aligned to the research questions: anomaly detection and fraud analytics, sequence modeling and transformer-based learning, credit risk and loan eligibility prediction, and explainability/counterfactual analysis. Sources were selected from peer-reviewed research, research-based books, and evidence-based industry frameworks. Search terms included financial fraud detection, autoencoder anomaly detection, transformer fraud detection, credit scoring machine learning, counterfactual explanations, explainable AI, model governance, and AI risk management. The sources were included if they directly supported model selection, evaluation metrics, interpretability, governance, or financial-risk context.

Table 2

Literature relevance matrix

Author (Year)	Domain/Context	Dataset/Setting	Methods	Key Findings	RQ Linkage

Goodfellow, Bengio, & Courville (2016)	Deep learning foundations	General neural network theory	Deep neural networks, optimization	Explains representation learning and optimization principles used in autoencoders and neural models.	Supports RQ1 model selection.
Vaswani et al. (2017)	Sequence modeling	Machine translation benchmark setting	Transformer/s elf-attention	Self-attention captures long-range dependencies more effectively than recurrent structures.	Supports transformer approach for RQ1.
Lundberg & Lee (2017)	Explainable AI	Model-agnostic interpretability	SHAP values	Provides a unified approach to feature attribution across models.	Supports RQ2/RQ4 explainability.

Wachter, Mittelstadt, & Russell (2017)	Counterfactual explanations	Algorithmic decisions	Counterfactual reasoning	Explains how minimal changes can make automated decisions understandable.	Supports RQ2/RQ4.
Ustun, Spangher, & Liu (2019)	Recourse in ML	Credit-like decision systems	Actionable recourse models	Shows how model explanations can become actionable recommendations.	Supports loan counterfactual recommendations.
Rudin (2019)	Interpretable ML	High-stakes domains	Interpretable vs black-box models	Argues that high-stakes domains require transparent or explainable models.	Supports governance rationale.

NIST (2023)	AI risk management	AI governance framework	Govern, Map, Measure, Manage	Defines trustworthy AI dimensions, including validity, reliability, transparency, and fairness.	Supports governance and compliance framing.
CFPB (2025)	Consumer finance data	Public consumer complaint database	Regulatory complaint data	Provides complaint data about financial products and services sent to companies for response.	Supports CFPB data source.
Chen (2025)	Financial fraud detection	Systematic review of fraud literature	Deep learning review	Summarizes recent deep learning architectures for financial fraud detection.	Supports RQ1 relevance.

Federal Reserve Payments Improvement (2021)	Synthetic identity fraud	Payment/fraud operations	Machine learning discussion	Notes machine learning can learn changing fraud patterns faster than static rules.	Supports fraud detection motivation.
---	--------------------------	--------------------------	-----------------------------	--	--------------------------------------

Thematic Literature Synthesis

The first theme is the shift from static rules to adaptive modeling. Fraud patterns change quickly, and static business rules often lag behind adversarial behavior. Deep learning methods, especially autoencoders, are appropriate when fraud labels are incomplete or imbalanced because they can model normal behavior and flag deviations through reconstruction error. This directly supports the RQ1 plan to compare deep learning approaches with simpler baselines.

The second theme is sequence and attention modeling. Transformer architectures introduced by Vaswani et al. (2017) demonstrate that attention mechanisms can capture dependencies across sequences. In financial settings, this concept is relevant because suspicious behavior may not appear in one isolated event; it may appear as a pattern across products, channels, issues, time, or complaint escalation behaviors. This supports the transformer-inspired component of the fraud/anomaly detection workflow.

The third theme is interpretability and actionable recourse. SHAP values, counterfactual explanations, and recourse frameworks move analysis from only predicting an outcome to explaining how the outcome might change. This is essential in financial decision systems because model predictions may affect consumers, investigators, credit officers, and compliance

reviewers. The notebooks demonstrate this concept by generating counterfactual recommendations for bad-credit cases and measuring whether predicted risk probabilities can be reduced.

The fourth theme is governance. The NIST AI Risk Management Framework emphasizes that trustworthy AI systems should be valid, reliable, safe, secure, transparent, accountable, and fair. In financial services, model performance alone is insufficient. This project therefore connects technical evaluation to fairness metrics, counterfactual transparency, and auditability.

Data Description

Data Source(s) and Access

Two primary data streams are used. The loan eligibility analysis uses the UCI German Credit dataset, which is a commonly used benchmark for credit risk classification. The fraud/anomaly risk workflow uses the CFPB Consumer Complaint Database, a public dataset containing complaints about financial products and services that are sent to companies for response. The project repository provides notebook code and paths for reproducing the analysis.

GitHub repository: https://github.com/KiranKBobba/Fin_Risk_Analytics.git

Data sources: UCI German Credit dataset; CFPB Consumer Complaint Database; FRED macroeconomic credit indicators used in the first notebook.

Dataset Overview

Table 3

Dataset overview

Dataset/Stage	Rows	Variables	Unit of Analysis	Target/Notes
Loan eligibility dataset	1,000 records	43 columns after feature enrichment	Borrower/application record	Target variable: binary credit outcome; notebook prints 700 bad and 300 good due to target coding used in the experiment.
Training split	800 records	29 scaled features after selected preprocessing	Training observation	Stratified split preserved class distribution: 560 target=1 and 240 target=0.
Test split	200 records	29 scaled features	Holdout observation	Test distribution: 140 target=1 and 60 target=0.
PCA representation	800 train / 200 test	21 PCA components	Reduced-dimensional record	21 components retained approximately 95.33% of

				explained variance.
Fraud/anomaly dataset	10K to 25K randomized CFPB records planned	Selected 7 to 10 complaint behavior variables	Consumer complaint record	Target variable is engineered fraud/anomaly risk label derived from issue keywords and operational indicators.

Data Dictionary

Table 4

Data dictionary

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
target	Binary credit outcome used for loan	Integer	0/1	No missing values	Model target in loan notebook.

	eligibility modeling				
duration	Loan duration in months	Numeric	4 to 72 months observed	No missing values	Core credit risk feature.
credit_amount	Loan amount	Numeric	250 to 18,424 observed	No missing values	Used in risk and counterfactual analysis.
age	Applicant age	Numeric	19 to 75 observed	No missing values	Used directly and in interaction features.
installment_rate	Installment burden category	Numeric/ordinal	1 to 4	No missing values	Risk burden indicator.
loan_to_duration	Credit amount divided by duration	Engineered numeric	Continuous	Derived	Captures intensity of loan amount per month.
loan_to_age	Credit amount	Engineered numeric	Continuous	Derived	Captures credit amount

	divided by age				relative to borrower age.
installment_burden	Credit amount multiplied by installment rate	Engineered numeric	Continuous	Derived	Captures repayment pressure.
duration_x_amount	Interaction between duration and amount	Engineered numeric	Continuous	Derived	Major counterfactual recommendati on variable.
age_x_employment	Interaction between age and employment encoding	Engineered numeric	Continuous	Derived	Captures stability proxy.
Product	Financial product in CFPB	Categorical	CFPB product categories	Unknown/blank handled by imputation	Fraud/anomaly feature.

	complaint data				
Issue	Consumer issue in CFPB complaint data	Categorical	CFPB issue categories	Unknown/blank handled by imputation	Primary fraud/anomaly risk signal.
Submitted via	Complaint submission channel	Categorical	Web, phone, referral, etc.	Unknown/blank handled by imputation	Operational behavior signal.
Timely response?	Whether company responded in time	Categorical	Yes/No	Unknown/blank handled by imputation	Operational anomaly indicator.
Consumer disputed?	Whether consumer disputed response	Categorical	Yes/No or missing depending on data version	Unknown/blank handled by imputation	Risk escalation indicator.

GitHub Data Availability Statement

The raw and processed data paths, model code, and notebook evidence are maintained under the GitHub repository. The intended repository structure is: data/raw for source extracts,

data/processed for cleaned modeling files, notebooks for executable analysis, outputs/figures for charts and figures, and docs for interim and final reports.

Analysis

Data Cleaning

The loan notebook reports no missing values for the original German Credit variables, but macro-FRED variables were unavailable for the synthetic monthly merge and therefore appeared as missing across the 1,000 records. These macro values were imputed using forward-fill and fallback zero logic in the notebook. Categorical features were encoded, numerical features were standardized, and PCA was used to reduce dimensionality. In the fraud/anomaly workflow, public CFPB data required categorical null handling and careful engineered label validation because first-row chronological sampling produced a single-class target. The current correction is to use randomized mixed-issue sampling and broader fraud/anomaly risk indicators.

Table 5

Data cleaning log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing macro values	FRED credit indicators	Missing value count after merge	Forward-fill, then fill remaining with zero	The synthetic time index did not align perfectly with all FRED periods.

Categorical values	German Credit categorical variables	Data type review	Label encoding	XGBoost and PCA require numeric inputs.
Scale differences	Credit amount, duration, age, engineered ratios	Feature scale review	StandardScaler normalization	Prevents large-scale variables from dominating PCA/modeling.
Dimensionality	29 scaled model features	PCA explained variance	PCA retained 21 components for 95.33% variance	Improves efficiency while preserving information.
CFPB label imbalance	fraud_label	Target distribution check	Randomized mixed-issue sampling and broader keyword labeling planned	Avoids invalid single-class model training.
Duplicates and outliers	Notebooks/credit data	EDA and descriptive review	Outlier retention for credit amount and duration	Extreme values may be meaningful in risk modeling.

EDA Results (Interpretation Emphasis)

The EDA figures generated in the notebooks identify several important patterns. First, the target variable is imbalanced, with 70% of cases coded as target=1 and 30% coded as target=0 in the loan eligibility experiment. Second, credit amount and loan duration display visible

dispersion and outliers. Third, engineered variables such as loan-to-age and installment burden show meaningful correlation with the target. These results justify using nonlinear machine learning models and counterfactual analysis rather than relying only on simple thresholds.

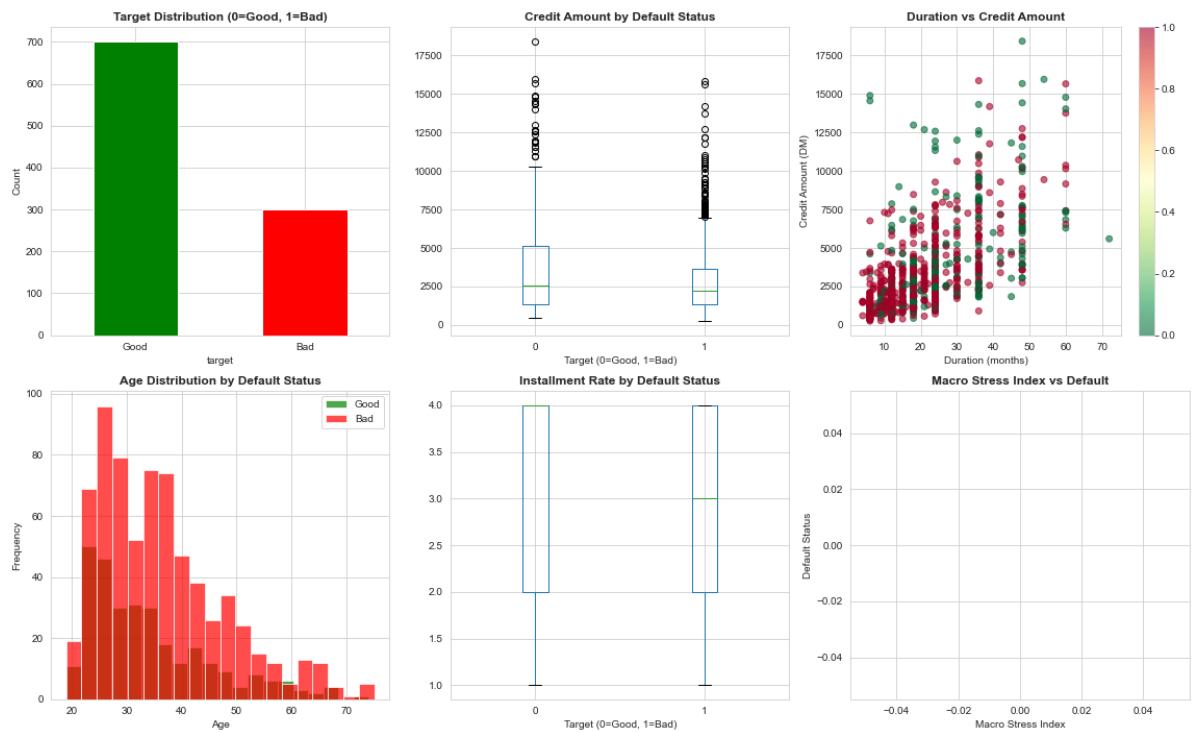


Figure 1. EDA panel showing target distribution, credit amount patterns, duration versus amount, age distribution, installment rate behavior, and macro-stress placeholder behavior.

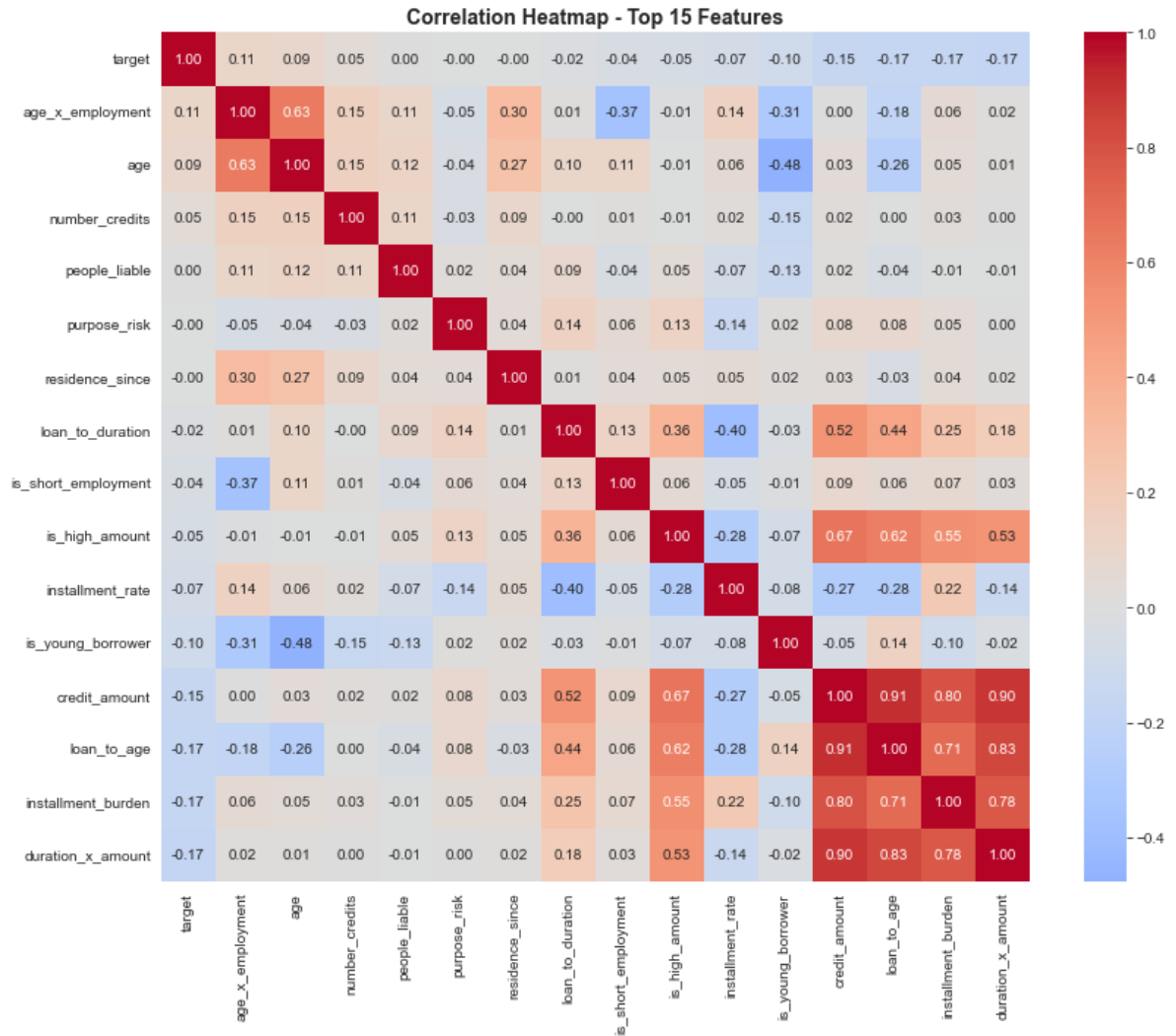


Figure 2. Correlation heatmap from the loan eligibility notebook. The analysis indicates that engineered variables such as loan-to-age, installment burden, and credit amount are more informative than many raw variables.

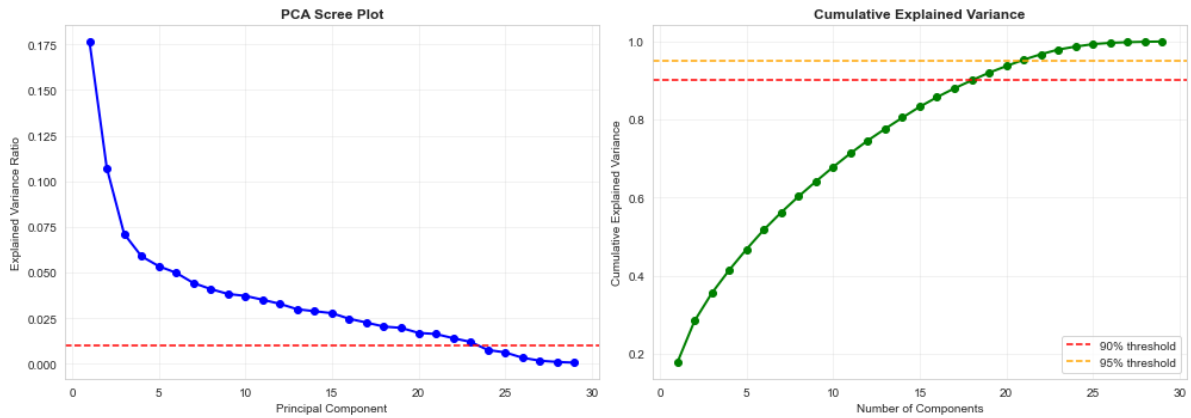


Figure 3. PCA explained variance plot. Twenty-one components retained approximately 95.33% of the variance, reducing feature dimensionality while preserving most information.

Table 6

EDA insight summary

Figure/Table Reference	What it Shows	Key Insight	Why it Matters	Decision/Next Step
Figure 1	Target and borrower feature distributions	The target distribution is imbalanced; credit amount and duration show wide variation.	Class imbalance and financial dispersion affect model evaluation and threshold selection.	Use stratified split and evaluate F1/ROC-AUC, not accuracy alone.
Figure 2	Correlation structure	Engineered variables show stronger	Supports feature engineering	Retain engineered behavior variables for modeling and

		relationships to the target than several raw features.	decisions for RQ3 and RQ4.	counterfactual analysis.
Figure 3	PCA explained variance	21 components retain 95.33% variance from 29 scaled features.	Addresses efficiency and multicollinearity.	Compare PCA versus original scaled feature performance.
Notebook output	Class distribution	Training set: 560 target=1 and 240 target=0; test set: 140 target=1 and 60 target=0.	Confirms stratification.	Use recall, specificity, and confusion matrix analysis.
Counterfactual output	Risk reduction patterns	Average probability improvement was 36.50%; success rate was 10/10 among	Supports RQ4 transparency and recourse.	Validate with statistical tests and business feasibility review.

		analyzed bad credits.		
--	--	--------------------------	--	--

Modelling

Choice of Models with Justification

Rule-based baseline: The rule-based baseline uses simple thresholds for credit amount, installment burden, loan-to-age, duration, and age. It represents a traditional explainable approach comparable to static business rules used in financial settings.

XGBoost classifier: XGBoost was selected because it handles nonlinear relationships, mixed feature interactions, and tabular risk data effectively. It is a strong baseline for structured credit risk and loan eligibility prediction.

PCA-enhanced XGBoost: The PCA version tests whether dimensionality reduction improves efficiency while preserving predictive signal. It also helps manage correlated engineered variables.

Autoencoder: Autoencoders are appropriate for fraud/anomaly detection because they can learn normal behavior and flag deviations via reconstruction error. This supports RQ1.

Transformer-inspired model: A transformer-inspired approach is relevant because attention mechanisms can model sequence-like dependencies across financial behaviors or complaint patterns. This remains in progress for final validation.

Counterfactual analysis: Counterfactual analysis was selected because it provides actionable explanations by showing what minimal feature changes would alter a prediction. This directly supports RQ2 and RQ4.

Features Included and Feature Engineering

Table 7

Feature set and usage

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
duration	Original	Numeric	Loan duration influences repayment risk and exposure length.	Rule baseline, XGBoost, PCA, counterfactual
credit_amount	Original	Numeric	Large loan amounts increase exposure and are central to underwriting.	Rule baseline, XGBoost, PCA, counterfactual
age	Original	Numeric	Borrower age may proxy stability and experience.	XGBoost, PCA, counterfactual
installment_rate	Original	Ordinal/numeric	Repayment burden category.	Rule baseline, XGBoost, PCA

loan_to_duration	Engineered	Numeric ratio	Captures loan intensity by month.	XGBoost, PCA, counterfactual
loan_to_age	Engineered	Numeric ratio	Captures amount relative to borrower age.	Rule baseline, XGBoost, PCA
installment_burden	Engineered	Numeric interaction	Combines loan amount and installment burden.	Rule baseline, XGBoost, counterfactual
duration_x_amount	Engineered	Numeric interaction	Captures joint exposure of amount and duration.	Counterfactual analysis
Product	Original CFPB	Categorical	Captures financial product risk context.	Fraud/anomaly workflow
Issue	Original CFPB	Categorical	Primary complaint behavior signal.	Fraud/anomaly workflow

Timely response?	Original CFPB	Categorical	Operational anomaly indicator.	Fraud/anomaly workflow
Consumer disputed?	Original CFPB	Categorical	Escalation/dispute signal.	Fraud/anomaly workflow

Evaluation Metrics and Formulae

Table 8

Evaluation metrics and formulae

Metric	Formula	Interpretation	Used For
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correctness; useful but insufficient under imbalance.	Baseline model comparison.
Precision	$TP / (TP + FP)$	Share of predicted positive risk cases that are truly positive.	Important when false positives are costly.
Recall	$TP / (TP + FN)$	Share of actual positive risk cases captured by the model.	Critical for fraud/risk detection.
F1-score	$2 * Precision * Recall / (Precision + Recall)$	Balances precision and recall.	Primary comparison under class imbalance.

ROC-AUC	Area under ROC curve	Measures ranking ability across thresholds.	Model discrimination.
Specificity	$TN / (TN + FP)$	Ability to correctly identify low-risk cases.	Fairness and business cost review.
Paired t-test	$t = \text{mean}(\text{differences}) / \text{SE}(\text{differences})$	Tests whether counterfactual probability reductions differ from baseline.	RQ4 statistical evidence.
Disparate impact	Positive rate unprivileged / positive rate privileged	Fairness comparison across groups.	Governance evaluation.

Preliminary Results

Preliminary Model Performance

Table 9

Preliminary model performance comparison

Model	Feature Set	Validation Approach	Metric 1	Metric 2	Key Takeaway
Rule-based baseline	Hand-crafted thresholds	Holdout test set	Accuracy 0.4250	F1 0.3353; AUC 0.5702	Transparent but weak recall and

					overall performance.
XGBoost (PCA)	21 PCA components	Holdout test set	Accuracy 0.7150	F1 0.8068; AUC 0.7270	Substantially better than rule baseline.
XGBoost (No PCA)	29 scaled features	Holdout test set	Accuracy 0.7250	F1 0.8110; AUC 0.7468	Slightly stronger than PCA version but less compact.
XGBoost (PCA)	21 PCA components	5-fold cross-validation	Mean ROC-AUC 0.7104	Std 0.0476	Performance is moderately stable across folds.
Counterfactual module	Bad-credit cases	10 analyzed test cases	Success 10/10	Average probability improvement 36.50%	Strong interim evidence for explainability and recourse.

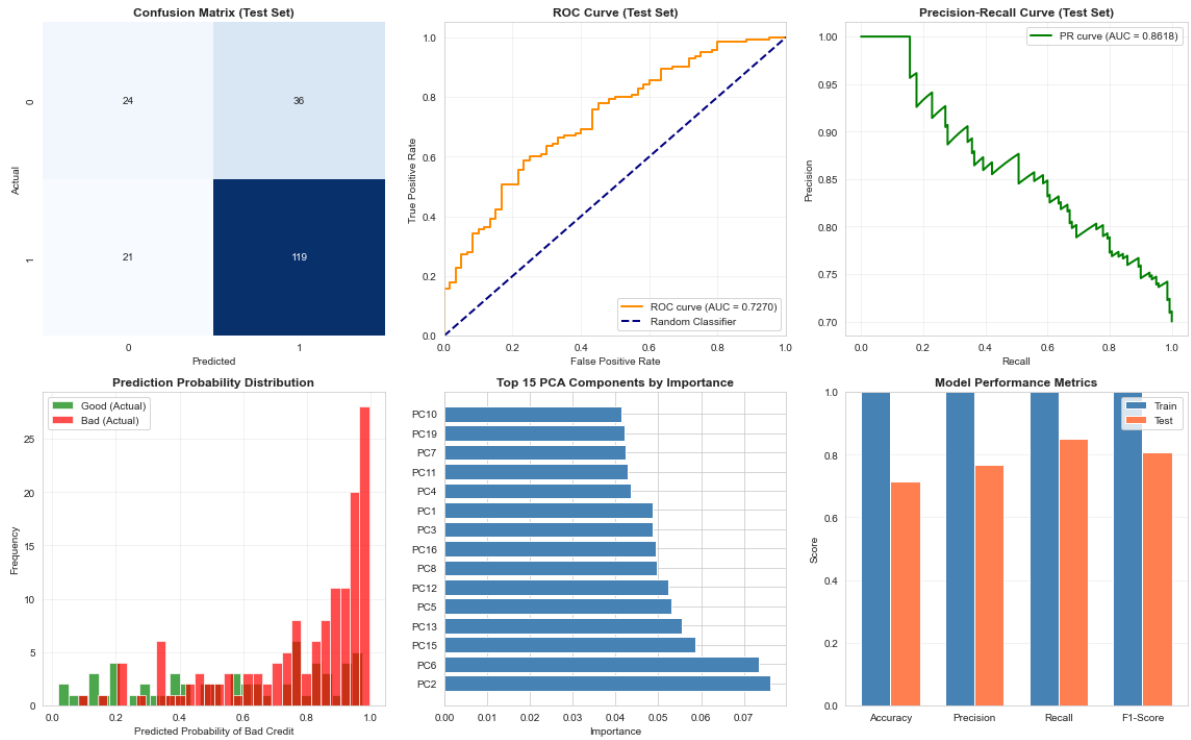


Figure 4. Model evaluation visualizations from the loan eligibility notebook, including confusion matrix, ROC/PR curves, metric comparison, and feature importance style diagnostics.

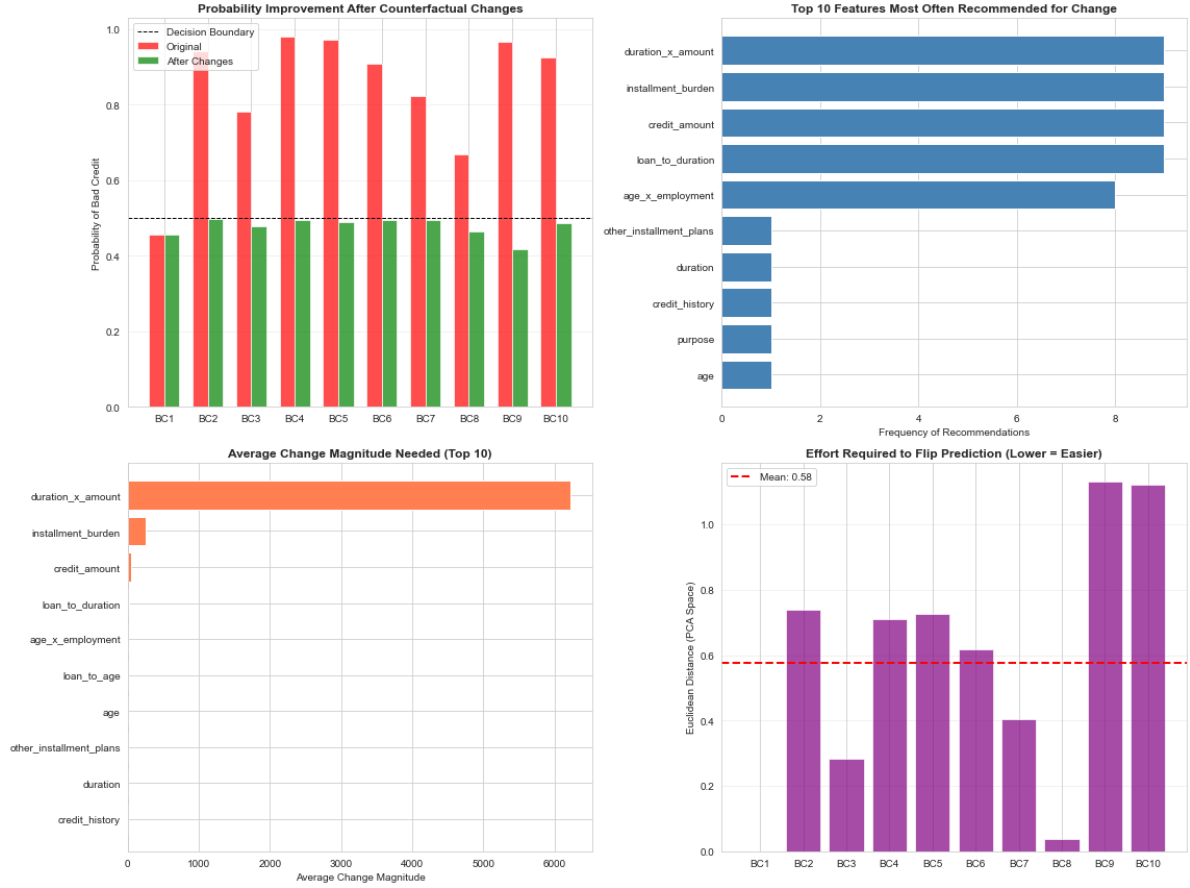


Figure 5. Counterfactual recommendation visualizations showing probability reductions and distance-to-flip analysis for selected bad-credit cases.

Preliminary Findings by Research Question

RQ1 preliminary finding: The fraud/anomaly workflow is still being corrected to ensure a valid mixed-class target from CFPB complaints. The interim conclusion is that direct model claims should wait until randomized mixed-issue sampling is complete. The model architecture and preprocessing design are prepared, and autoencoder/transformer approaches remain appropriate based on the literature and project objectives.

RQ2 preliminary finding: Counterfactual logic has already demonstrated value in the loan eligibility workflow. The counterfactual module reduced predicted bad-credit probability across

10 analyzed cases, with a reported average probability improvement of 36.50% and a 10/10 success rate in reaching below the decision threshold. This supports the proposition that counterfactual explanations improve interpretability by showing how predictions can change.

RQ3 preliminary finding: Binary classification for loan eligibility achieved useful predictive performance. XGBoost with PCA achieved 71.5% accuracy, 0.8068 F1-score, and 0.7270 ROC-AUC. XGBoost without PCA achieved slightly higher performance at 72.5% accuracy, 0.8110 F1-score, and 0.7468 ROC-AUC. The evidence supports rejecting the RQ3 null hypothesis at this interim stage, while noting that additional tuning is required.

RQ4 preliminary finding: Counterfactual reasoning improved transparency by identifying recommended changes such as `duration_x_amount`, `installment_burden`, `credit_amount`, `loan_to_duration`, and `age_x_employment`. The statistical tests in the notebook showed a paired t-test p-value of 0.0001 and Wilcoxon p-value of 0.0077, indicating that pre/post counterfactual probability changes were statistically meaningful in the analyzed sample.

Table 10

Research-question-specific interim conclusions

RQ	Focus	Evidence from Notebooks	Interim Decision
RQ1	Fraud/anomaly deep learning effectiveness	Fraud/anomaly target correction is still in progress. Architecture and data pipeline are prepared.	Do not make final hypothesis decision yet; continue with random mixed-issue 10K sample.

RQ2	Counterfactual fraud prediction explanation	Counterfactual logic identifies feature changes that shift model outcomes; full fraud-specific validation pending.	Preliminary support for Ha, pending final fraud/anomaly runs.
RQ3	Binary loan eligibility prediction	XGBoost outperformed rule baseline; F1 improved from 0.3353 to approximately 0.8068/0.8110.	Reject H0 and support Ha at interim stage.
RQ4	Counterfactual transparency and fairness	Counterfactual results showed 36.50% average probability improvement and significant pre/post tests.	Reject H0 and support Ha at interim stage, subject to fairness refinement.

Interpretability and Business Implications

The most important interpretability result is that the counterfactual analysis did not only classify borrowers; it produced actionable recommendations. Across 10 bad-credit cases, the notebook identified `duration_x_amount`, `installment_burden`, `credit_amount`, `loan_to_duration`, and `age_x_employment` as frequent recommendation variables. This means the model is sensitive to both financial behavior variables and engineered interactions that represent exposure, repayment burden, and stability proxies.

From a business perspective, these variables help translate technical outputs into underwriting questions. For example, credit amount and duration jointly reflect exposure; installment burden reflects payment pressure; loan-to-duration summarizes monthly intensity; and age-employment interaction approximates borrower stability. These are not merely

mathematical variables but operational indicators that loan officers and risk teams can evaluate. The fraud/anomaly workflow will use a similar approach by identifying complaint issue type, product, submission channel, dispute status, and timely-response behavior as interpretable risk drivers.

The counterfactual results also have governance value. A model that predicts 'high risk' without explanation is difficult to defend. A model that says risk would fall if specific measurable variables changed is more transparent, more auditable, and more useful for responsible decision support. However, final recommendations must avoid suggesting unrealistic or unfair changes. For example, age cannot be changed by the applicant, so age-related explanations should be treated as stability proxies or fairness audit signals rather than applicant instructions.

Interim Limitations and Risks

Table 11

Interim limitations and mitigation plan

Limitation/Risk	Current Observation	Planned Mitigation
Public data limitation	The CFPB dataset does not contain internal bank transaction fraud labels.	Define the target as fraud/anomaly risk rather than confirmed fraud.
Synthetic time index	German Credit data does not contain real timestamps.	Use synthetic dates only for experimental enrichment, not causal temporal claims.

Macro-feature missingness	FRED variables were missing across the synthetic merge in current run.	Either improve alignment or treat macro variables as exploratory context.
Overfitting risk	XGBoost training accuracy was 1.0000 while test accuracy was 0.7150.	Tune max_depth, regularization, learning rate, and use cross-validation.
Fairness risk	Protected or proxy variables may influence outcomes.	Use fairness metrics and recourse review before final recommendations.
Counterfactual realism	Some suggested changes may not be directly actionable.	Add feasibility constraints and human review to counterfactual generation.
Fraud target imbalance	Initial CFPB labeling generated only class 0 when sampled chronologically.	Use randomized mixed-issue sampling and validate target distribution before modeling.

Next Steps (for Final Report)

- Replace chronological CFPB sampling with randomized 10K mixed-issue sampling and verify that the engineered fraud/anomaly label contains both classes.
- Re-run fraud/anomaly feature importance and model comparison after valid label generation.
- Complete autoencoder fraud/anomaly detection with reconstruction error thresholding and compare against random forest/logistic baseline.

- Complete transformer-inspired fraud/anomaly model using reduced text length and memory-safe embedding parameters.
- Add SHAP explanations for the best-performing structured model.
- Add counterfactual examples for fraud/anomaly predictions and identify which financial behavior variables cause prediction changes.
- Tune XGBoost and compare PCA versus no-PCA models using cross-validation.
- Refine fairness analysis by segmenting outcomes across age/stability proxies or available demographic/context groups.
- Update GitHub repository with raw data paths, processed data, notebooks, figures, and final report artifacts.

Responsible AI and Governance Considerations

Because the project operates in a financial decision context, responsible AI concerns are central. Fraud detection can incorrectly flag legitimate customers, while loan eligibility models can deny credit or increase perceived risk. Therefore, evaluation must include both predictive metrics and governance metrics. This is why the interim report includes precision, recall, F1-score, ROC-AUC, disparate impact, equal opportunity difference, and statistical tests for counterfactual changes.

The NIST AI Risk Management Framework supports this approach by emphasizing governance, measurement, mapping of risks, and active management. In this project, governance means documenting the data source, explaining engineered targets, testing model performance, monitoring fairness, and ensuring counterfactual explanations are not misleading. The final project should therefore present both model results and limitations transparently.

The use of CFPB complaint data also requires careful wording. The dataset contains consumer complaints, not verified fraud transactions. Therefore, the correct term for that workflow is fraud/anomaly risk, not confirmed fraud. This distinction strengthens the academic quality of the study because it avoids overstating what the public data can prove.

Bibliography

Chen, Y. (2025). Deep learning in financial fraud detection: A systematic review. *Journal of Finance and Data Science*.

Consumer Financial Protection Bureau. (2025). Consumer Complaint Database.

<https://www.consumerfinance.gov/data-research/consumer-complaints/>

Federal Reserve Payments Improvement. (2021). Next-level detection through machine learning. Federal Reserve.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. U.S. Department of Commerce.

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1, 206-215.

UCI Machine Learning Repository. (1994). *Statlog (German Credit Data)*. University of California, Irvine.

Ustun, B., Spangher, A., & Liu, Y. (2019). Actionable recourse in linear classification. *Proceedings of the Conference on Fairness, Accountability, and Transparency*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*.

Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law and Technology*, 31(2), 841-887.

Appendix A: Notebook Evidence and Code Availability

The uploaded notebooks contain the executable evidence used for this interim report. The German Credit notebook includes steps for feature engineering, EDA, PCA, XGBoost modeling, rule-based baseline modeling, counterfactual analysis, and fairness tests. The 25K optimized notebook contains the fraud/anomaly risk extension and the updated label-engineering/final-conclusion sections. The code and data paths will be maintained in the GitHub repository listed at the beginning of the report.

Table 12

Notebook and repository evidence

Artifact	File/URL	Purpose
Notebook 1	AI_Driven_Financial_Risk_Analytics(2).ipynb	Loan eligibility, feature engineering, PCA, XGBoost, counterfactual analysis, fairness tests

Notebook 2	AI_Driven_Financial_Risk_Analytics_25K_FIXED.ipynb	Fraud/anomaly risk extension, memory-optimized data, enhanced fraud label logic, feature importance and conclusions
GitHub	https://github.com/KiranKBobba/Fin_Risk_Analytics.git	Repository for code, datasets, notebooks, and report artifacts

Appendix note: Figures in this report were extracted from the uploaded notebooks where available. Additional diagrams and workflow tables were created to align with the interim report template and to make the project status understandable as a standalone document.